Sheldon Liu

AI Full-Stack Software Engineer | Melbourne, Australia
+61 472 737 067 | lxd040715@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI full-stack software engineer who turns LLM capabilities into measurable, shippable product features, with end-to-end experience across Flutter and React / Next.js front ends, TypeScript, Deno and Python back ends, PostgreSQL, RAG retrieval and LLM integration. Recently built the AI assistant for a creative social platform end to end, from Flutter features to an evaluation-gated Supabase back end with grounded retrieval and test-driven delivery.

TECHNICAL SKILLS

Languages: TypeScript, Python, Dart, Kotlin, JavaScript, SQL, Java, C#, C
AI & RAG: LLM APIs (OpenAI, Claude, Gemini, OpenRouter), LLM routing and evaluation, RAG, vector search (pgvector, Qdrant), embeddings, Hugging Face, prompt engineering
Full-Stack & Mobile: Flutter, Next.js (App Router), React 19, HTML, CSS, Tailwind CSS, Node.js, Express, FastAPI, ASP.NET, Deno, REST APIs, SSE, Android (Jetpack Compose), JWT, OAuth 2.0
Data & Cloud: PostgreSQL, Supabase (Edge Functions, Row-Level Security), MongoDB, Prisma ORM, Upstash Redis, Trigger.dev, Docker, Vercel, Render, AWS, Cloudflare R2
Engineering: Test-driven development, unit and integration testing, pgTAP, Vitest, Git, GitHub Actions CI/CD

PROFESSIONAL EXPERIENCE

Ft.ured **Jul 2026 – Oct 2026**
AI Full-Stack Engineer (University Industry Project) | *Flutter, Dart, Deno, TypeScript, Supabase, PostgreSQL, pgvector, Row-Level Security, Edge Functions, Trigger.dev, OpenRouter*

- Designed the decision-routing layer for the client's AI assistant on a social app for creatives: an LLM classifies each request and routes it across five product capabilities, with cost and latency logged for every decision and row-level security protecting user data.
- Built a 241-message evaluation set to compare LLMs on routing accuracy, safety, latency and cost; improved routing accuracy from **82.5% to 97.4%**, with all prompt-injection tests passing at ~620 ms P95 latency.
- Developed natural-language People Discovery using hybrid keyword and pgvector search with an LLM relevance layer to filter and rank candidates; achieved **0.91 precision and 0.97 top-5 recall** under 2.4 s P95 on 50 unseen queries (3 runs each).
- Developed a grounded in-app Q&A assistant backed by a curated knowledge base, with navigation limited to approved screens; achieved a **96.4%** blind-evaluation pass rate across 211 questions with zero fabricated answers.
- Implemented Flutter features end to end, including multi-conversation chat, AI-assisted onboarding, job-post drafting and content suggestions, working test-first with **1,000+ back-end and 450+ Flutter tests** in CI.

SELECTED PROJECTS

ShopMate — RAG-based Conversational E-commerce Assistant (Android) **2026**
Independent Developer | *Kotlin, Jetpack Compose, Node.js, TypeScript, Express, PostgreSQL, Qdrant, OpenAI API, SSE, Render*

- Built an Android shopping assistant end to end, from the Kotlin app to the Node.js back end, database and LLM integration, turning plain-language requests into product recommendations, Q&A, comparisons, cart updates and a simulated checkout.
- Combined Qdrant vector search with PostgreSQL lookups, keeping product data and cart validation in the database rather than the LLM; reached **1.000 Recall@5** and 0.989 MRR@10 on a fixed 111-query offline baseline.
- Implemented SSE streaming of AI responses and prompt compression, cutting **first-token latency by about 60%**; added request tracing and logging to tune retrieval; deployed on Render.

DevStash — Full-Stack Knowledge Management SaaS for Developers **2026**
Independent Developer | *Next.js 16, React 19, TypeScript, Prisma 7, Neon PostgreSQL, Auth.js, Stripe, Cloudflare R2, Upstash Redis*

- Built a SaaS platform for storing, searching and editing code snippets, prompts, commands, notes, files, images and links.
- Integrated LLM APIs for automatic tagging, summaries and code explanations; implemented the data model, Pro-plan feature gating and Stripe subscription billing.
- Implemented rate limiting for authentication and AI features with Upstash Redis, covered by Vitest tests.

EDUCATION

University of Melbourne **Mar 2024 – Mar 2027**
Bachelor of Science, Major in Computer Science | *Melbourne, Australia*

- Relevant coursework: Algorithms and Data Structures, Software Modelling, Database Systems, Foundations of Artificial Intelligence, Network Computing, IT Project.